$10^{-27}$ erg-second is Planck's quantum mechanical constant, and $c = 2.998 \times 10^{10}$ centimeter/second is the speed of light.

**plasma**. Hot, ionized, electrically conducting gas.

**plutonium-239.** A specific type of plutonium atomic nucleus which contains 239 protons and neutrons (94 protons and 145 neutrons).

**polarization.** The property that electromagnetic and gravitational waves have of consisting of two components, one that oscillates in one direction or set of directions, and the other in a different direction or set of directions. The two components are called the waves' two polarizations.

**polarized body.** A body with negative electric charge concentrated in one region and positive charge concentrated in another region.

**polarized light; polarized gravitational waves.** Light or gravitational waves in which one of the two polarizations is completely absent (vanishes).

**postdoc.** Postdoctoral fellow; a person who has recently received the Ph.D. degree and is continuing his or her training in how to do research, usually under the guidance of a more senior researcher.

**pressure.** The amount of outward force that matter produces when it is squeezed.

**Price's theorem.** The theorem that all properties of a black hole that can be converted into radiation will be converted into radiation and will be radiated away completely, thereby making the hole "hairless."

**primordial black hole.** A black hole typically far less massive than the Sun that was created in the big bang.

**principle of absoluteness of the speed of light.** Einstein's principle that the speed of light is a universal constant, the same in all directions and the same in every inertial reference frame, independent of the frame's motion.

**principle of equivalence.** The principle that in a local inertial reference frame in the presence of gravity, all the laws of physics should take the same form as they do in an inertial reference frame in the absence of gravity.

**principle of relativity.** Einstein's principle that the laws of physics should not be able to distinguish one inertial reference frame from another; that is, that they should take on the same form in every inertial